

CHEM 279 Final Report : CNDO/2 Based Calculation of Polarity and Polarizability for Molecular Classification

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Abstract

This project extends a self-consistent CNDO/2 molecular electronic structure framework to calculate molecular dipole moments and polarizability tensors under externally applied electric fields. Dipole moments were computed using both an approximate Mulliken charge method and a more rigorous dipole integral approach, which were then compared against experimental literature values. Finite central differences were used to determine the polarizability tensor, from which isotropic and anisotropic polarizabilities were derived. While the CNDO/2 calculations systematically underestimated polarizabilities due to the limitations of the semiempirical method and minimal STO-3G basis set, the overall qualitative trends in molecular polarity and polarizability were reproduced successfully. These results demonstrate that CNDO/2 can serve as an efficient first-pass molecular classification tool for identifying candidate materials with desirable electronic properties.

1 Background

Understanding how molecules respond to external electric fields is an important problem in chemistry because these responses govern many observable chemical and physical properties. Two of the most significant electronic response properties are the permanent electric dipole moment and the molecular polarizability tensor, both of which describe how charge is distributed within a molecule and how that distribution changes when an external electric field is applied. These quantities are directly related to intermolecular interactions, dielectric behavior, spectroscopic response, refractive properties, and molecular alignment in electric fields, all of which play vital roles in the design of materials systems. As a result, computational prediction of these properties is valuable in a wide variety of fields, including materials science, semiconductor fabrication, analytical chemistry, and molecular simulation, among many others.

From the dipole moment and polarizability tensor we can extract a number of molecular properties that describe how the desired molecule reacts to electric field stimuli. The permanent dipole magnitude, calculated as the magnitude of the dipole vector, measures the overall separation of positive and negative charge in a molecule at equilibrium and quantifies the molecule's polarity. Similarly, the isotropic polarizability and anisotropic polarizability can be calculated from the elements of the polarizability tensor. The isotropic polarizability, calculated as the average of the diagonal elements of the polarizability tensor, measures how easily a molecule's electron cloud can be uniformly distorted in all directions by an external field while the anisotropic polarizability, calculated as the deviation from this isotropic average, describes how directional the response to the external electric field is.

There are many industrial applications for these derived electronic molecular properties. The permanent dipole magnitude is used to predict solvent behavior, dielectric constants, intermolecular binding strength, and spectroscopic signatures. Likewise, isotropic polarizability can be used to estimate van der Waals interactions as well as the refractive index and optical response, which is important for modeling liquids and photonic materials. Anisotropic polarizability expands upon isotropic polarizability by providing additional information about the directional dependence of this optical and electronic behavior, making it relevant to nonlinear optics and liquid crystal systems.

Because these electronic properties strongly influence the behavior of experimentally observable materials, they can serve as useful descriptors for identifying molecules with desirable functional characteristics. By computationally calculating these properties, this work intends to provide a first-pass screening tool for selecting molecular candidates for polar solvents, dielectric materials, and systems requiring a strong directional or optical response.

2 Objectives

This project aims to extend the self-consistent CNDO/2 model of the wavefunction to extract both the electric dipole moment and electric polarizability of a wide variety of molecules when subjected to an external electric field. Two methods of extracting the dipole moment of the molecules of interest are explored: an atom-centered approximation method using Mulliken charges, and an exact calculation via dipole integrals. These two approaches are then evaluated on their accuracy by comparisons to experimental values found within literature. Once the dipole moment has been determined, we then apply small perturbations in the strength of the external electric field to approximate the polarizability tensor via central differences. From the dipole moment and polarizability tensors, we can extract the permanent dipole moment, isotropic polarizability, and anisotropic polarizability, all of which collectively describe how the molecule interacts with an external electric field. After implementing this property calculation, our final goal is to rank the molecules passed into the classification pipeline by their values of these properties. We can then use these classifications to identify promising candidates for certain desired material applications, such as solubility in a polar solution, and dielectric and optical responses.

We chose to explore a wide variety of molecules, ranging in symmetries, polarities, size, and number of pi-bonds to explore how these factors affect a molecule’s electronic properties. For example, a symmetric molecule may have no permanent dipole due to individual bond dipoles canceling out due to geometry. Furthermore, the number of lone pairs and electronegativities of the individual atoms within a molecule may affect the strength of the overall dipole moment. Similarly, the size of a given molecule plays a defining role, as larger molecules have more electrons that are distributed over a larger volume, making them easier to distort when exposed to an external electric field. Finally, the number of pi-bonds within a given molecule may increase the overall polarizability as well, as pi-electrons are bound less tightly than electrons in a sigma bond and typically lie above and or below the bond axis. Each of these factors contribute to a molecule’s overall polarity and polarizability, however, within combinations of these factors it is unclear as to which trend may dominate, which this project seeks to define.

The molecules of interest for this project are as follows, with detailed information on their symmetry, polarity, and number of pi-bonds found in Appendix B.

Table 1: Molecules of Interest

Compound	Formula
Methane	CH ₄
Ethane	C ₂ H ₆
Propane	C ₃ H ₈
Acetylene	C ₂ H ₂
Ethylene	C ₂ H ₄
Benzene	C ₆ H ₆
Carbon Dioxide	CO ₂
Water	H ₂ O
Ammonia	NH ₃
Hydrogen Fluoride	HF

3 Methodology

In order to develop our molecular classification pipeline, it was necessary to compute two properties for a given molecule, the dipole moment and the polarizability tensor. Conceptually, the dipole moment and the polarizability are respectively the first and second derivatives of the energy with respect to a change in external electric field. Using these quantities we can compute several derived properties, including dipole strength, isotropic polarizability, and anisotropic polarizability. These quantities are used in the molecular classification pipeline, as they provide tangible information about molecular properties.

3.1 Dipole Moment

There are several reasonable accurate approximations that can be used to calculate the dipole moment with CNDO/2 theory. Here, we explore the Mulliken approximation, as well as the full dipole integral method over atomic orbitals.

3.1.1 Dipole Approximation

The Mulliken approximation is atom-centered, and assumes the electron cloud for each atom is centered on the nucleus. We can use the optimized density matrix, $P_{\mu\nu}$, from the CNDO/2 SCF cycle, to calculate this. The total density matrix is the sum of the α and β contributions from unrestricted Hartree Fock.

$$P_{\mu\nu} = P_{\mu\nu}^{\alpha} + P_{\mu\nu}^{\beta}$$

Using this matrix, we can calculate the net electron population (N_A) on atom A as the sum of the density matrix on that atom.

$$N_A = \sum_{\mu \in A} P_{\mu\mu}$$

We can then calculate the net atomic charge on each atom (q_A) as the difference between the effective nuclear charge (Z_A) and the net electronic population.

$$q_A = Z_A - N_A$$

The dipole moment is classically a function of charge and position. Therefore, our net dipole moment (μ) is the following.

$$\mu = \sum_A q_A \mathbf{R}_A$$

3.1.2 Dipole Integral

The dipole integral method is a more robust way to calculate the dipole moment for CNDO/2. More detailed theory behind the calculation of the dipole integral is included in Appendix D. Using this method, the net formula we use to compute the x component of the dipole moment is the following.

$$\mu_x = -\frac{\partial E}{\partial F_x} = -\sum_{\mu\nu} P_{\mu\nu} D_{\mu\nu}^x - \sum_A Z_A \mathbf{R}_{A,x}$$

Here, $P_{\mu\nu}$ is the density matrix, $D_{\mu\nu}^x$ is the dipole integral matrix, Z_A is the effective nuclear charge, and $R_{A,x}$ is the x component of the location of atom A . The formula for $D_{\mu\nu}^x$ is defined in the detailed theory section. Each cartesian component of the dipole is computed in a similar fashion, to yield the full dipole vector, $\vec{\mu}$.

$$\vec{\mu} = \langle \mu_x, \mu_y, \mu_z \rangle$$

3.2 Polarizability

The polarizability is conceptually the second derivative of the energy with respect to the change in external electric field. We label this quantity α , where $\alpha = \frac{\partial^2 E}{\partial F^2}$. Given that it is significantly more difficult to derive a closed-form integral solution for this, we are using the central differences theorem to approximate the polarizability. This has the following theoretical formula, where i and j both represent cartesian directions.

$$\alpha_{ij} = \left(\frac{\partial \mu_i}{\partial F_j} \right)_{\mathbf{F}=0}$$

Using the central differences approximation, we can evaluate this as the following.

$$\alpha_{ij} \approx \frac{\mu_i(+F_j) - \mu_i(-F_j)}{2F}$$

In order to evaluate each α_{ij} term, we use our existing implementation of the CNDO/2 self-consistent field cycle. This requires being able to compute the dipole moment as a function of the electric field. To accomplish this, we run our SCF cycle as normal, but with a field-dependent perturbation added to the hamiltonian. This quantity, $\Delta h_{\mu\nu}$, is a function of the both electric field and dipole integral terms ($D_{\mu\nu}^x$), which takes on the following form.

$$\Delta h_{\mu\nu} = F_x D_{\mu\nu}^x + F_y D_{\mu\nu}^y + F_z D_{\mu\nu}^z$$

We perform this calculation for all combinations of direction j for electric field perturbation (F_j) and resulting changes in the dipole vector in direction i (μ_i). This will yield a 3×3 matrix for each combination of i and j , which represents the following polarizability tensor.

$$\alpha = \begin{pmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{pmatrix}$$

3.3 Derived Properties

We compute three scalar derived properties from the dipole ($\vec{\mu}$) and polarizability tensor (α). These include the dipole strength, isotropic polarizability, and anisotropic polarizability. They are calculated as follows.

3.3.1 Dipole Strength (μ)

The dipole strength (μ) is computed as the magnitude of the dipole vector ($\vec{\mu}$). This represents the permanent polarity of the structurally optimized molecule.

$$\mu = |\vec{\mu}|$$

3.3.2 Isotropic Polarizability ($\bar{\alpha}$)

The isotropic polarizability is computed from the polarizability tensor. It represents the induced polarity of a molecule when an external electric field is applied.

$$\bar{\alpha} = \frac{\alpha_{xx} + \alpha_{yy} + \alpha_{zz}}{3}$$

3.3.3 Anisotropic Polarizability ($\Delta\alpha$)

The anisotropic polarizability is also computed from the polarizability tensor. It quantifies how directional the induced dipole response to the external electric field is.

$$\Delta\alpha = \left[\frac{(\alpha_{xx} - \alpha_{yy})^2 + (\alpha_{yy} - \alpha_{zz})^2 + (\alpha_{zz} - \alpha_{xx})^2 + 6(\alpha_{xy}^2 + \alpha_{yz}^2 + \alpha_{xz}^2)}{2} \right]^{\frac{1}{2}}$$

4 Results and Discussion

After running our molecular classification pipeline for our molecules of interest, we obtain the following results, comparing both the approximate method for obtaining the dipole magnitude with the more rigorous dipole integral, both of which are compared to experimental values (Figure 1).

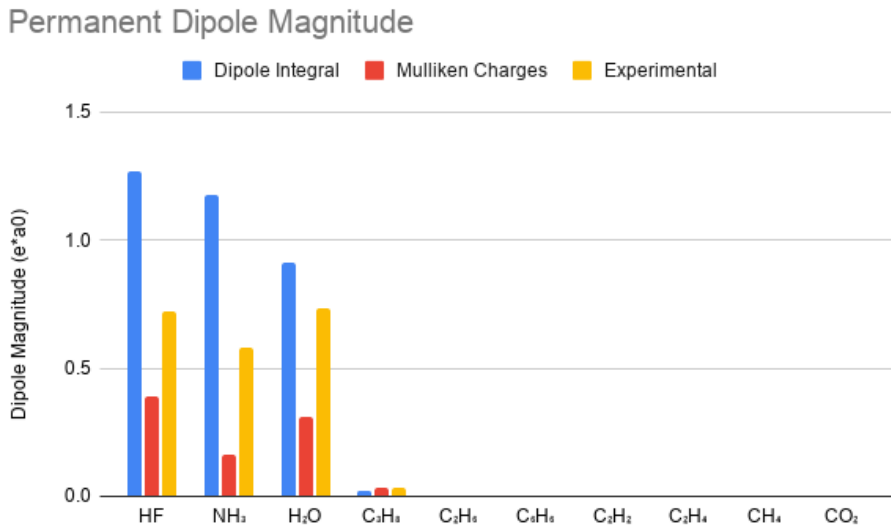


Figure 1: Comparison of permanent dipole magnitude between approximated method, dipole integration, and experimental values[1]. See Appendix C for values.

As expected, the polar molecules have much larger dipole moments, with the nonpolar molecules essentially having no dipole moment at all. This is a result of the geometry of each molecule, as symmetric, or near symmetric, molecules have their individual bond dipole moments nullified by one another. Within the polar molecules, we would expect that as the number of lone pairs and electronegativities of the atoms increase, the overall magnitude of the dipole moment would increase as well. However, this was not the case for the results for water. Experimentally, water has the greatest dipole magnitude as compared to the other polar molecules we investigated (HF, NH₃). Since oxygen is more electronegative than nitrogen, and oxygen contains an additional lone pair within water as compared to nitrogen within ammonia, we would anticipate the dipole magnitude of water to be greater, but our results did not reflect this. Furthermore, while the dipole integration method overestimates the permanent dipole magnitude, the approximated method underestimates it, further demonstrating this inaccuracy. This is a result from the semi-empirical nature of the CNDO/2 model, but nonetheless, the overall trend is consistent between polar and nonpolar molecules.

Isotropic Polarizability

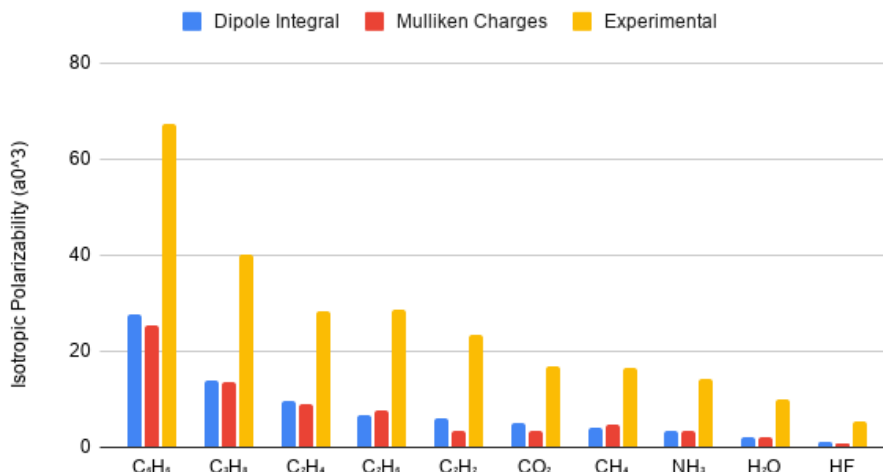


Figure 2: Comparison of isotropic polarizability between approximated method, dipole integration, and experimental values[1]. See Appendix C for values.

For isotropic polarizability, we can see that the main trend dominating this value is size, since as the number of carbon atoms within the molecule increases, so does its isotropic polarizability. However, an interesting nuance lies within the values for the two carbon molecules (C_2H_4 , C_2H_6 , C_2H_2), with ethylene having the largest isotropic polarizability of these molecules, then ethane, and finally acetylene. Initially, we expected that the polarizability would increase with the number of pi-bonds. However, it seems that the geometry of the molecule is a greater overall factor. Since isotropic polarizability is the average of the polarizabilities in all three cartesian directions, if a molecule’s geometry is highly linear, as it is in acetylene, then while polarizability will be greater along the specific bond direction, the other two directions will be much smaller. This will cause the overall polarizability average to decrease, leading to smaller isotropic polarizabilities.

In comparison to the experimental values, our results match the overall trend well. However, quantitatively, both the approximate Mulliken charge and the exact dipole integration methods result in isotropic polarizabilities that are much lower than the experimental values. Once again, this is due to the limitations of the semi-empirical CNDO/2 theory and our choice of a minimal STO-3G basis set representation. Because polarizability depends on how easily the electron cloud can distort under an external electric field, the limited flexibility of the basis functions restricts the magnitude of the induced electronic response. This restriction is best seen in the larger molecules with greater delocalization of their pi-electrons, where the disparity between the experimental and predicted values is at its greatest. Although the calculated magnitudes are underestimated, both methods of extracting the isotropic polarizability still reproduce the expected qualitative trends with respect to molecular size, symmetry, and number of pi-bonds.

Anisotropic Polarizability

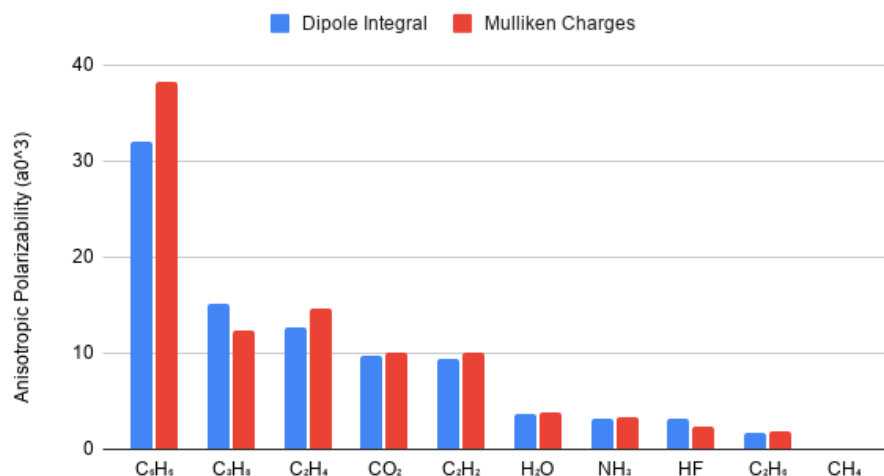


Figure 3: Comparison of anisotropic polarizability between approximated method, dipole integration, and experimental values. See Appendix C for values. No experimental literature values were able to be located for anisotropic polarizability.

From Figure 3, overall anisotropic polarizability increases when a molecule has a strong directional preference for electron distortion, such as in linear, elongated, or pi-bonded systems. Small symmetric molecules like methane and ethane have low anisotropy because their electron clouds respond similarly in all directions. Molecules like benzene show much larger anisotropy because their pi-electrons and larger size allow stronger polarization along certain molecular axes. Molecular geometry plays an important role in anisotropic polarizability, as linear molecules like carbon dioxide show much larger polarizability in their pi-electrons along their bond direction than in other directions. This leads to larger differences in polarizability when comparing different Cartesian directions, resulting in anisotropic polarizabilities that are much higher than their isotropic counterparts. In general, anisotropic polarizability reflects molecular shape and directional electron delocalization rather than just overall molecular size.

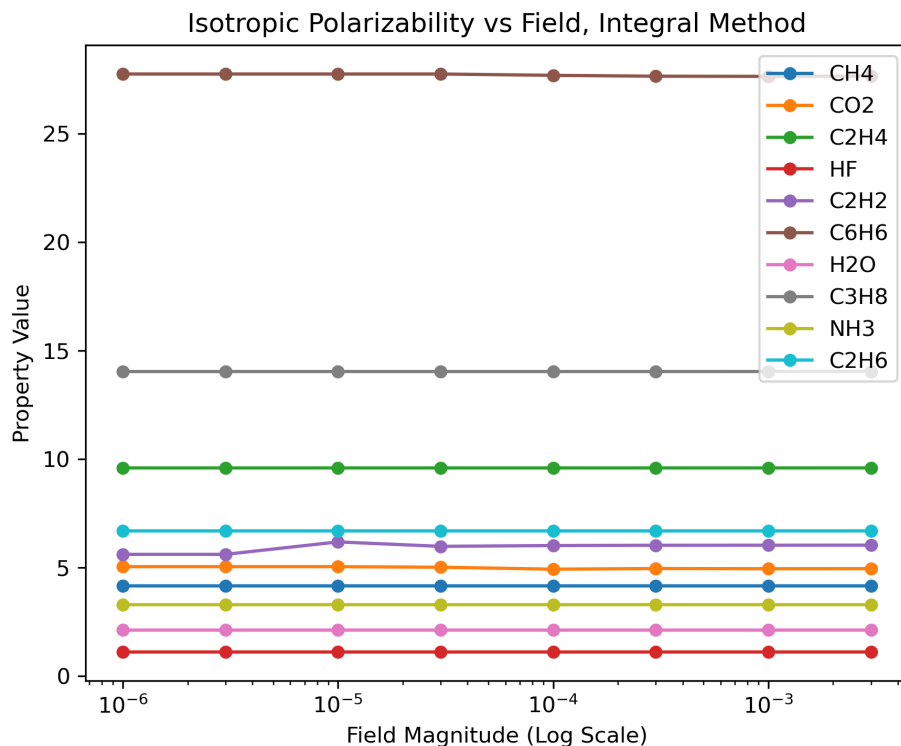


Figure 4: Isotropic Polarizability at varying External Electric Field Strength, Integral Method.

To ensure physical consistency, a plot of the isotropic polarizability as a function of the strength of the external electric field was generated in Figure 4, correctly indicating that these values remain mostly constant regardless of the field strength. The full set of plots for isotropic and anisotropic polarizability, computed using both dipole calculation methods, are included in Appendix F. This validates that our model is indeed calculating these values correctly, as isotropic and anisotropic polarizabilities are intrinsic molecular properties that are affected by molecular size, symmetry, and delocalization of electrons. Polarizability is a linear response of the dipole moment to an applied electric field, meaning that for small perturbations in this field, the dipole changes proportionally to the field strength. As a result, the slope of the dipole response (i.e. the polarizability) should remain constant, which Figure 4 confirms.

Polarizability Anisotropy Ratio

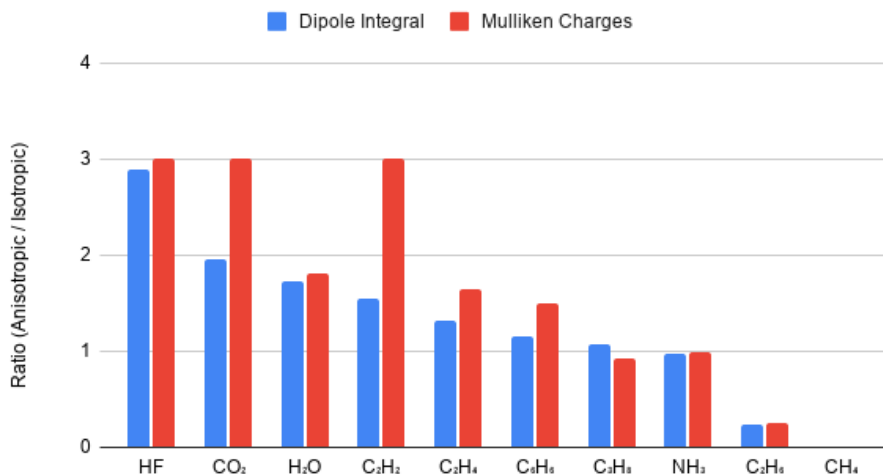


Figure 5: Polarizability Anisotropy Ratio for the Molecules of Interest. See Appendix C for values.

The polarizability anisotropy ratio in Figure 5 provides a normalized measure of how directional a molecule’s electronic response is, but unlike anisotropic polarizability, is independent of molecule size, allowing for greater insight into the molecule’s underlying geometry and distribution of electrons. Highly symmetric molecules such as methane and ethane exhibit small anisotropy ratios because their electron density distorts similarly in all directions, while linear or asymmetric molecules such as HF, CO₂, and C₂H₂ exhibit larger ratios due to stronger distortions along their molecular axis and weaker distortions in other directions. Similarly, molecules containing pi-bond delocalization such as C₂H₄ and benzene display greater directional response because their pi-electrons are bound less tightly, making them easier to displace by an external electric field.

5 Conclusions

Taking into consideration the limitations of our approach, these results are what we might reasonably expect from a CNDO/2 based molecular property determination code. As discussed, the absolute values of the properties do not perfectly match experimental values, but on the whole, the trend compares very well. We would expect this given the imprecision resulting from neglecting differential overlap, and the limitations of the STO-3G basis set. If we were to implement methods which are closer to ab initio, this discrepancy would likely be improved. However, given that the purpose of developing a molecular classification pipeline is to produce correct orderings rather than numerically precise values, CNDO/2 is an excellent choice for this application, as it produces accurate orderings while being significantly less computationally expensive than true ab initio methods.

These limitations are further mitigated by the intended use case of the pipeline, which is to identify a handful of promising candidates with various material applications. For example, a very reasonable workflow could be to use the CNDO/2 based method as an efficient first pass to select the top 10% of candidates, from thousands, which have a given material property. From there, more computationally expensive ab initio methods such as DFT could be used to further down-select these candidates to those that should be investigated experimentally. This approach also mitigates the impact of minor imperfections in the CNDO/2 ordering; as long as general trends are accurate, the most promising candidates should still make this first cut. While it is not common to use CNDO/2 methods in industry given that significantly more accurate methods exist, this would be a legitimate application of the theory in a way that could be valuable.

While our pipeline has so far only been applied to a handful of molecules, we can still identify the top candidates which would have certain desired material properties. The ordered candidates are included

in the table in Appendix E. The first classification identifies polar candidates, which are ranked by the value of the permanent dipole. This can be used to determine molecules which are soluble in polar media, or those that can act as a polar interface. The second classification identifies dielectric candidates, which are ranked by large values of both permanent dipole and isotropic polarizability. These materials will have a strong induced polar response to an external electric field, which can be useful for applications such as capacitors, printed circuit boards, or cable insulators. The third classification identifies optical response candidates, which are ranked by polarizability. Excited state methods are typically better for this purpose, but this criteria is a strong indicator of materials which will be responsive to light, and therefore can be used to identify chromophores, or materials with refractive responses. The final classification identifies directional response candidates, which are ranked by anisotropic polarizability. This type of material would be composed of molecules which tend to align in an electric field. These molecules can help create pathways for ion transport in batteries, act as liquid crystals in devices such as transistors, or even behave as molecular machines.

6 Future Work

There are many natural extensions that can be applied to this project. To start, a more diverse set of molecules can be explored with even larger sizes and greater delocalization of electrons. CNDO/2 theory works best for smaller molecules, so it would be interesting to see at what molecule size the qualitative trends begin to break down. Likewise, we could test more complex basis sets to see if the quantitative values for the molecular properties that we derived can better match the experimental results.

Additionally, we could seek to extract another electronic molecular property, the hyperpolarizability, which describes the nonlinear response of a molecule to an applied electric field. While polarizability measures the linear change in the dipole moment with respect to the electric field, hyperpolarizability measures how this response itself changes as the field strength increases. It is a natural progression to this project, as dipole moment is the first derivative of molecular energy with respect to an external electric field, polarizability is the second derivative, and hyperpolarizability is the third derivative. Very asymmetrical molecules as well as those with extended electron delocalization would be especially interesting for this application due to their enhanced nonlinear optical response.

7 Contributions

Christian Fernandez was responsible for selecting the molecules of interest for this project as well as ensuring their input files in the code repository were physically accurate. He implemented the basic skeleton for the code repository, as well as functions for extracting the derived molecular properties once the dipole vector and polarizability tensor were found. In this report he was responsible for the Background, Objectives, Results / Discussion, and Future Work sections.

Alison Jarvis was responsible for the underlying theory of extracting the dipole vector and polarizability tensor from our existing CNDO/2 code as well as implementing them into the code repository. She created the main molecular classification pipeline as well as tracked how the derived molecular properties varied with electric field strength. In this report, she was responsible for the Abstract, Methodology, and Conclusion sections as well as aiding in creating graphs for the Results / Discussion section.

8 Appendix

A Github Repository Link

Repository Link: <https://github.com/alison-jarvis/chem279-final-project.git>

B Molecules of Interest Properties Table

Table 2: Chemical Properties of Common Compounds

Name	Symbol	Symmetric?	Polarity	Number of π -bonds
Methane	CH ₄	✓	Nonpolar	0
Ethane	C ₂ H ₆	✓	Nonpolar	0
Propane	C ₃ H ₈		Nonpolar	0
Acetylene	C ₂ H ₂	✓	Nonpolar	2
Ethylene	C ₂ H ₄	✓	Nonpolar	1
Benzene	C ₆ H ₆	✓	Nonpolar	3
Carbon Dioxide	CO ₂	✓	Nonpolar	2
Water	H ₂ O		Polar	0
Ammonia	NH ₃		Polar	0
Hydrogen Fluoride	HF		Polar	0

C Tables for Figures in Results/Discussions

Table 3: Permanent Dipole Magnitude

	Dipole Integral	Mulliken Charges	Experimental[1]
HF	1.26613	0.387589	0.719
NH ₃	1.17792	0.162901	0.581
H ₂ O	0.912104	0.307701	0.731
C ₃ H ₈	0.0227874	0.0353226	0.033
C ₂ H ₆	0.000661645	4.31×10^{-5}	0
C ₆ H ₆	2.23×10^{-14}	2.28×10^{-14}	0
C ₂ H ₂	1.71×10^{-14}	1.20×10^{-14}	0
C ₂ H ₄	1.11×10^{-14}	3.88×10^{-15}	0
CH ₄	6.24×10^{-15}	3.47×10^{-15}	0
CO ₂	1.39×10^{-15}	7.77×10^{-15}	0

Table 4: Isotropic Polarizability

	Dipole Integral	Mulliken Charges	Experimental[1]
C ₆ H ₆	27.7005	25.5308	67.21
C ₃ H ₈	14.0452	13.4174	39.96
C ₂ H ₄	9.60144	8.94481	28.26
C ₂ H ₆	6.69783	7.65925	28.52
C ₂ H ₂	6.02323	3.33087	23.53
CO ₂	4.93219	3.34753	16.92
CH ₄	4.1629	4.7397	16.52
NH ₃	3.29552	3.44131	14.19
H ₂ O	2.1244	2.07869	10.13
HF	1.11482	0.79073	5.4

Table 5: Anisotropic Polarizability

	Dipole Integral	Mulliken Charges
C ₆ H ₆	32.0959	38.2962
C ₃ H ₈	15.1021	12.4206
C ₂ H ₄	12.6566	14.65
CO ₂	9.67991	10.0426
C ₂ H ₂	9.35966	9.9926
H ₂ O	3.68417	3.77087
NH ₃	3.22462	3.43116
HF	3.21984	2.37219
C ₂ H ₆	1.63118	1.89138
CH ₄	5.94×10^{-11}	2.77×10^{-11}

Table 6: Polarizability Anisotropy Ratio

	Dipole Integral	Mulliken Charges
HF	2.888215138	3
CO ₂	1.962598764	3.000002987
H ₂ O	1.73421672	1.814060779
C ₂ H ₂	1.553927046	2.999996998
C ₂ H ₄	1.318198104	1.637821262
C ₆ H ₆	1.158675836	1.5
C ₃ H ₈	1.075249907	0.925708409
NH ₃	0.978485945	0.997050542
C ₂ H ₆	0.243538579	0.246940627
CH ₄	1.43×10^{-11}	5.84×10^{-12}

D Detailed Dipole Theory

To determine how to compute the dipole using the integral method we start from first principles, using the concept that the dipole is the first derivative of the energy with respect to a change in external electric field, which we call F . We know that the formula for the CNDO/2 energy is the following.

$$E_{CNDO/2} = \sum_{\mu\nu} P_{\mu\nu}(h_{\mu\nu} + f_{\mu\nu}) + E_{nuc}$$

The fock matrix, $f_{\mu\nu}$, is a function of $h_{\mu\nu}$, and additional terms which do not depend on the external electric field. Therefore, we consider the fock-specific additive contribution to be constant in the derivative. Using this concept and the fact that the density matrix itself is independent of an external electric field assuming converged SCF, we get the energy derivative as the following. Here, we are simplifying the derivation be for only the x component, but the form will be identical for the other components.

$$\frac{\partial E_{CNDO/2}}{\partial F_x} = \sum_{\mu\nu} P_{\mu\nu} \frac{\partial h_{\mu\nu}}{\partial F_x} + \frac{\partial E_{nuc}}{\partial F_x}$$

To determine $\frac{\partial h_{\mu\nu}}{\partial F_x}$, we need to understand how the hamiltonian depends on an external electric field. Below is the general notation for the hamiltonian as a function of F .

$$\hat{H}(F) = \hat{H}_0 + F_x \hat{x} + F_y \hat{y} + F_z \hat{z}$$

Again, simplifying to just the x component, if we take the derivative of this with respect to F_x , we get the following.

$$\frac{\partial \hat{H}}{\partial F_x} = \hat{x}$$

Therefore, the derivative of the hamiltonian is only a function of the position operator. Putting this into the atomic orbital notation used for CNDO/2, we get the following, where we refer to the $\frac{\partial h_{\mu\nu}}{\partial F_x}$ as $D_{\mu\nu}^x$ for simplicity.

$$D_{\mu\nu}^x = \langle \chi_\mu | x | \chi_\nu \rangle$$

Here, χ_μ and χ_ν are the atomic orbital basis functions and x represents the position operator. Given that we are specifically using the STO-3G basis set, which is comprised of contracted gaussians, the equation we use to compute $D_{\mu\nu}^x$ is as follows.

$$D_{\mu\nu}^x = \sum_k \sum_l d_{k\mu} d_{l\nu} N_{k\mu} N_{l\nu} D_x^{kl}$$

Here, k and l index the primitive gaussians in the basis set. The $d_{k\mu}$ and the $d_{l\nu}$ terms are the contraction coefficients of primitive gaussians k and l , respectively. Likewise, $N_{k\mu}$ and $N_{l\nu}$ are the normalization coefficients for these gaussians. Finally, D_{kl} represents the dipole overlap integral for two primitive gaussians. For two arbitrary primitive gaussians A and B , the dipole integral will be computed as follows.

$$D_x^{AB} = \int g_A(r) \cdot x \cdot g_B(r) dr$$

Using the same techniques as for our gaussian overlap integral, we can separate this into multiplicative components, to get the following.

$$D_x^{AB} = D_x^{AB,x} S_y^{AB} S_z^{AB}$$

The S^{AB} terms are our overlap integrals for y and z . In order to determine the $D_x^{AB,x}$ term, we need to express this as an integral over two primitive one dimensional gaussians. The equations for two arbitrary primitive 1D gaussians, A and B , are included below.

$$g_A = (x - X_A)^{l_A} (y - Y_A)^{m_A} (z - Z_A)^{n_A} e^{\alpha|r-R_A|^2}$$

$$g_B = (x - X_B)^{l_B} (y - Y_B)^{m_B} (z - Z_B)^{n_B} e^{\beta|r-R_B|^2}$$

Using this formulation, we get that the equation for the $D_x^{AB,x}$ term is as follows.

$$D_x^{AB,x} = \int (x - X_A)^{l_A} x (x - X_B)^{l_B} e^{-\alpha(x-X_A)^2 - \beta(x-X_B)^2} dx$$

After working out this integral, we can express this in terms of overlap integral terms, to get the following equation.

$$D_x^{AB,x} = S_x^{AB} (l_A + 1, l_B) + X_A S_x^{AB} (l_A, l_B)$$

Here, l_A and l_B refer to the angular momentum of primitive gaussians A and B , respectively, and X_A refers to the x coordinate of the center of primitive gaussian A . S_x^{AB} is the overlap integral term for x . We use these principles to calculate the $D_{\mu\nu}$ terms within our CNDO/2 code.

Going back to our overall calculate of the dipole moment, the nuclear energy expressed as a function of F is the following.

$$E_{nuc} = \sum_A Z_A (F_x R_{A,x} + F_y R_{A,y} + F_z R_{A,z})$$

Here, Z_A is the effective nuclear charge for a given atom. Therefore, the derivative of the nuclear energy with respect to a change in F_x works out to be the following.

$$\frac{\partial E_{nuc}}{\partial F_x} = \sum_A Z_A \mathbf{R}_{A,x}$$

We can then substitute this equation into our expression for a single component of the dipole, μ_x , to get the following.

$$\mu_x = -\frac{\partial E}{\partial F_x} = -\sum_{\mu\nu} P_{\mu\nu} D_{\mu\nu}^x - \sum_A Z_A \mathbf{R}_{A,x}$$

This is the complete equation we use to evaluate the dipole for each cartesian component.

E Material Property Candidates

Table 7: Top Three Candidates by Property

Category	Top 3 (Ordered)
Polar Candidates ($\uparrow \mu$)	HF, NH ₃ , H ₂ O
Dielectric Candidates ($\uparrow \mu$, $\uparrow \bar{\alpha}$)	C ₆ H ₆ , C ₃ H ₈ , NH ₃
Optical Response Candidates ($\uparrow \bar{\alpha}$)	C ₆ H ₆ , C ₃ H ₈ , C ₂ H ₄
Directional Response Candidates ($\uparrow \Delta\alpha$)	C ₆ H ₆ , C ₃ H ₈ , C ₂ H ₄

F Polarizability Invariability with Field

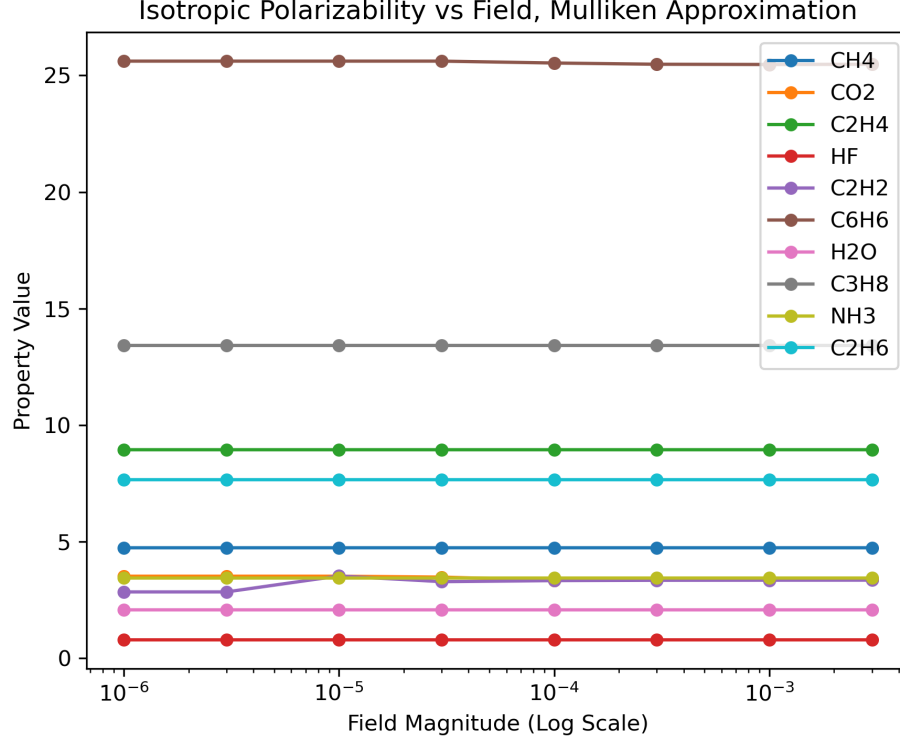


Figure 6: Isotropic Polarizability at varying External Electric Field Strength, Mulliken Method.

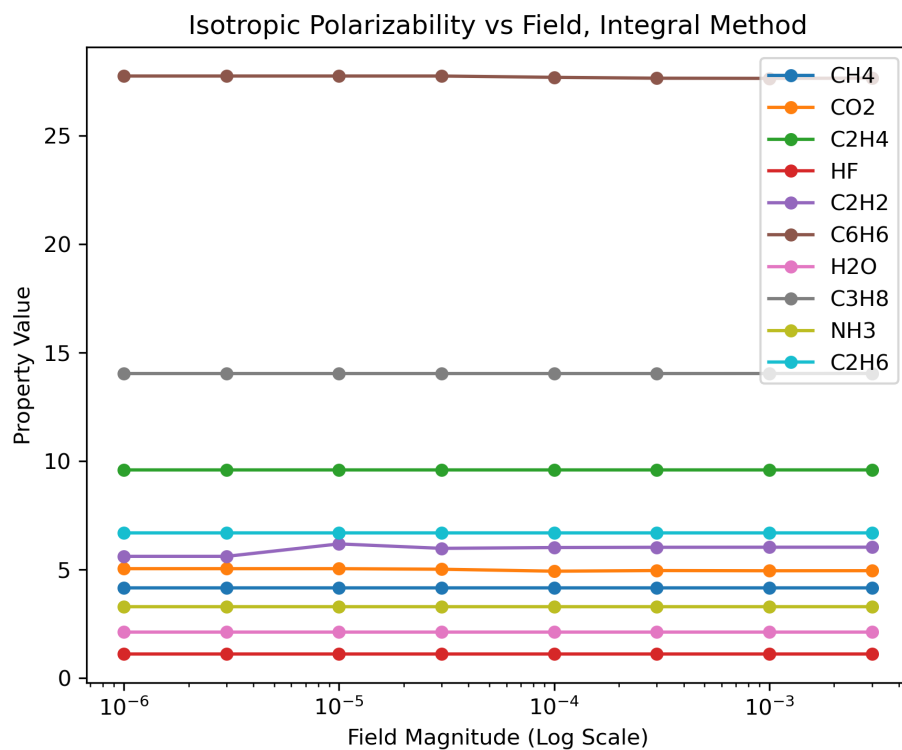


Figure 7: Isotropic Polarizability at varying External Electric Field Strength, Integral Method.

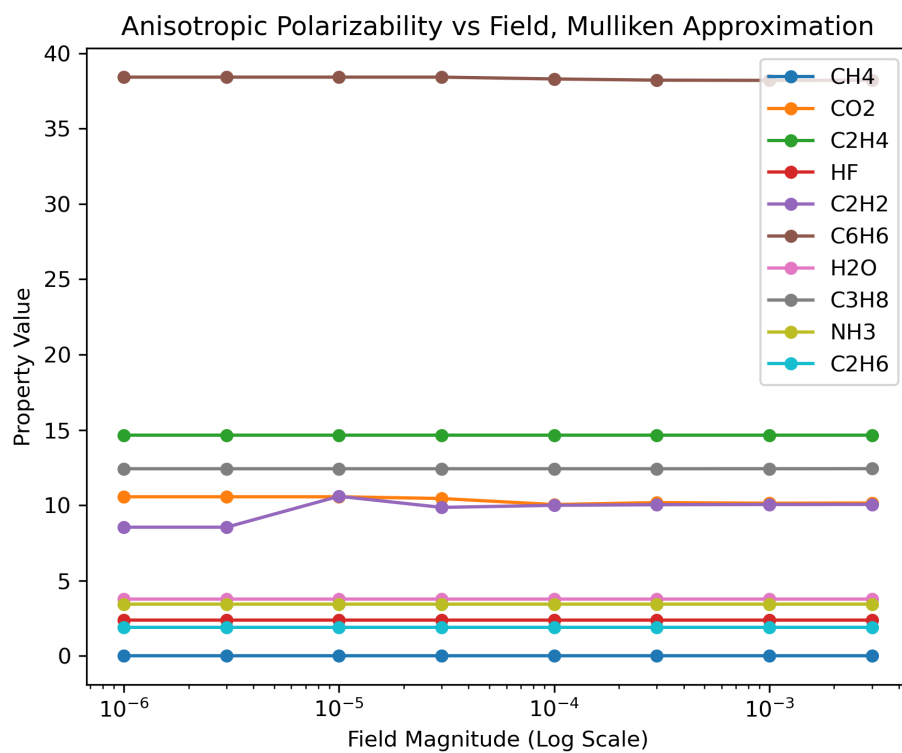


Figure 8: Anisotropic Polarizability at varying External Electric Field Strength, Mulliken Method.

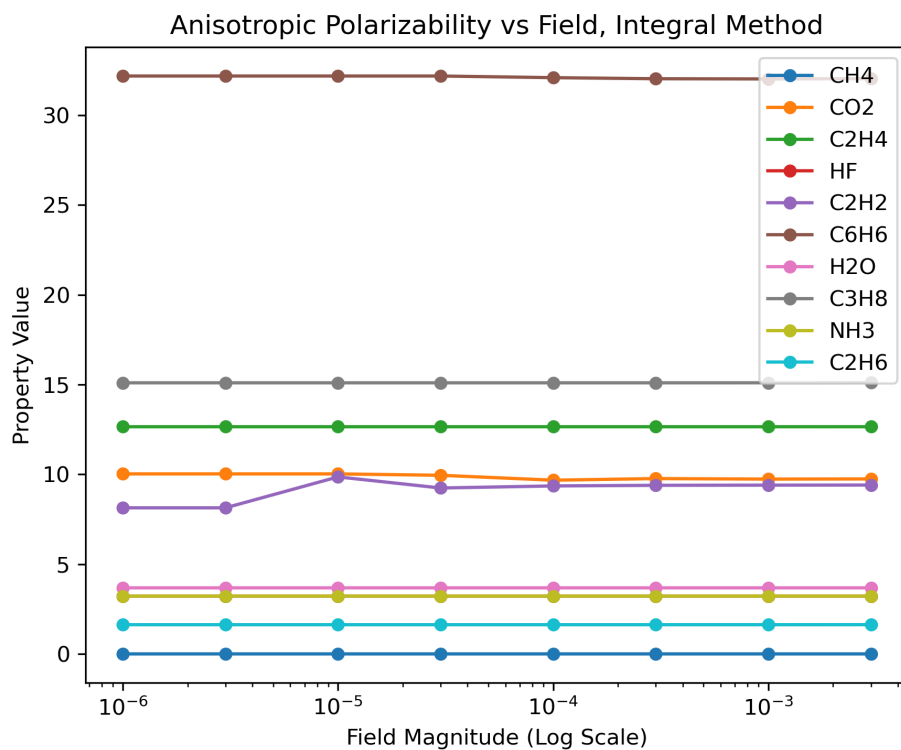


Figure 9: Anisotropic Polarizability at varying External Electric Field Strength, Integral Method.

References

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